Daniel A. Delgado, PhD

Software Engineer | Spatial Computing, AI & Product Development
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INDEPENDENT SOFTWARE PROJECTS

Galaxy Viz | Software Engineer & Data Visualization Developer | 2026–Present
Demo: xdaedalusx.github.io/galaxy-viz | GitHub: github.com/xdaedalusx/galaxy-viz
• Developed an interactive 3D visualization of academic papers using Three.js/WeabGL, featuring citation-based sizing, year-based coloring, citation and similarity networks, search, and timeline controls.
• Built a TypeScript data pipeline for BibTeX/CSV/JSON ingestion, TF-IDF, clustering, 3D layout, automated testing, and synthetic-data generation.
• Designed analytical features identifying disconnected research topics ("Voids") and conflicting findings ("Nebulas"), validated against synthetic ground truth; deployed on GitHub Pages.

iCarus | Software Engineer & UI Developer | 2026–Present
GitHub: github.com/xdaedalusx/icarus
• Developed a native macOS media player in SwiftUI with audio/video playback, downloads, and customizable interfaces.
• Engineered a modular skinning system supporting custom layouts and Windows Media Player (.wmz) import, including a sandboxed JavaScript runtime for skin animations and events.

SPATIAL COMPUTING & AI PROJECTS | UNIVERSITY OF FLORIDA

Collaborative AR Platform & Shared-Gaze Visualization | Software Engineer | 2021–2025
• Designed and developed a multi-user augmented reality application in Unity and C#, using Microsoft MRTK2 and HoloLens 2 eye tracking to support real-time collaborative assembly and sorting tasks.
• Implemented multiplayer networking with Photon Unity Networking (PUN), including synchronized object states, player avatars, scene management, remote procedure calls, and object ownership handling.
• Integrated Azure Spatial Anchors for cross-device spatial alignment and developed interactive gaze visualizations, hand-tracking interactions, physics, and collision systems.
• Built event-driven logging to capture timestamped gaze coordinates, user interactions, and task performance, with Python/Pandas scripts for automated log processing and behavioral analysis.

EMG-Based Prosthetic Control in VR | Software Engineer & Researcher | 2020–2023
• Developed Unity simulations in C#/.NET for VR-based prosthetic training, integrating real-time EMG control using Delsys sensors and Leap Motion hand tracking.
• Processed multi-channel EMG data with MATLAB and Python (NumPy, Pandas, scikit-learn) for signal filtering, feature extraction, and LDA-based gesture classification.
• Implemented feedforward and NARX neural networks in MATLAB for continuous prosthetic movement prediction and built a Python TCP socket server to transmit control signals to the Unity simulation.

EDUCATION

Ph.D., Computer Science — University of Florida, Gainesville, FL | 2025

TECHNICAL SKILLS

Programming & Web: Swift, SwiftUI, C#/.NET, Python, Java, JavaScript, TypeScript, React, Node.js, HTML, CSS, Git
Spatial Computing: Unity, AR/VR, HoloLens 2, eye tracking, computer vision
AI & Machine Learning: Deep learning, neural networks, TensorFlow, regression, EMG classification
Data Visualization: Three.js, WebGL, GLSL shaders, 3D visualization, network graphs, JSON
Product Development: UI architecture, interaction design, Figma, prototyping, responsive web development
Testing & AI Tools: Usability testing, experimental design, mixed-methods analysis, Claude, AI-assisted development